

A Counterexample to TLMC Conjecture 00000000548: $\iota(I) \leq \beta_0(I)$ Fails for Principal Monomial Ideals

TLMC falsification package

October 2, 2026

Abstract

TLMC conjecture 00000000548 asserts that for a monomial ideal I , the minimal number $\iota(I)$ of components in an irredundant irreducible decomposition is bounded by $\beta_0(I)$, the minimal number of monomial generators, claiming the bound is combinatorially verified for monomial ideals up to $n = 12$ with tightness attained by complete intersections. We exhibit the principal monomial ideal $(xyz) \subseteq k[x, y, z]$, for which

$$(xyz) = (x) \cap (y) \cap (z), \quad \iota((xyz)) = 3 > 1 = \beta_0((xyz)),$$

refuting the conjecture at $n = 3$ — inside the claimed verification range — by an ideal that is itself a complete intersection. All numbers are independently recomputed by exhaustive enumeration (`reproduce.py`) and formalized in axiom-free Lean 4 (`lean4/`).

1 The conjecture

00000000548. *Definition: the minimal number $\iota(I)$ of irreducible decompositions is the least number of components. Conjecture: $\iota(I) \leq \beta_0(I)$; and the combinatorial control of the decomposition number is verified up to $n = 12$ for monomial ideals, with tightness attained by complete intersections.*

Throughout, $R = k[x_1, \dots, x_n]$ and all ideals are monomial ideals, identified with their sets of exponent vectors $e \in \mathbb{N}^n$.

Definition 1. An *irreducible* monomial ideal is one generated by pure powers of a subset of the variables, $(x_{i_1}^{a_1}, \dots, x_{i_s}^{a_s})$. An *irreducible decomposition* of I is an expression $I = \bigcap_j J_j$ with all J_j irreducible; it is *irredundant* if no component can be dropped. $\iota(I)$ is the least number of components of an irredundant irreducible decomposition of I .

Definition 2. $\beta_0(I)$ is the cardinality of the unique minimal monomial generating set of I (the number of generators of I , i.e. the rank of $\text{Tor}_0(k, I)$).

2 The counterexample

Work in $R = k[x, y, z]$ and let $I = (xyz)$. Write $e = (e_x, e_y, e_z)$ for exponent vectors, and let $|_1, |_2, |_3$ denote the coordinates. The ideal generated by a monomial $x^a y^b z^c$ is the up-set $\{e : e_x \geq a, e_y \geq b, e_z \geq c\}$; the irreducible ideal (x^a, y^b, z^c) is $\{e : e_x \geq a \text{ or } e_y \geq b \text{ or } e_z \geq c\}$ (absent variables carry no constraint). Both descriptions are componentwise, so all equalities and inequalities of monomial sets verified inside a finite exponent box lift verbatim to R .

Theorem 3. $(xyz) = (x) \cap (y) \cap (z)$.

Proof. xyz divides a monomial $x^{e_x}y^{e_y}z^{e_z}$ iff $e_x \geq 1$ and $e_y \geq 1$ and $e_z \geq 1$, iff the monomial lies in each of (x) , (y) , (z) . $\square$

Lemma 4. (x) , (y) , (z) are irreducible monomial ideals. In fact, no two monomial ideals strictly containing (x) intersect to (x) , and similarly for (y) and (z) .

Proof. Suppose P and Q are monomial ideals with $P \supsetneq (x)$ and $Q \supsetneq (x)$. Choose $u \in P$, $v \in Q$ with $u_x = v_x = 0$ (such monomials exist because P, Q strictly contain (x) , which consists exactly of the monomials with x -exponent ≥ 1). Then $w := uv \in P$ and $w \in Q$ since both are up-sets, while $w_x = u_x + v_x = 0 < 1$, so $w \notin (x)$. Hence $P \cap Q \supsetneq (x)$. The cases of (y) and (z) are symmetric. $\square$

Lemma 5. The decomposition of Theorem 3 is irredundant.

Proof. $yz \in (y) \cap (z)$ but $yz \notin (xyz)$, so dropping (x) strictly enlarges the intersection; xz and xy play the same role for (y) and (z) . $\square$

Theorem 6. $\iota((xyz)) = 3$.

Proof. By Theorem 3 and Lemmas 4 and 5 there is an irredundant irreducible decomposition with 3 components, so $\iota \leq 3$. Conversely, irreducible monomial ideals are exactly the pure-power-generated ones, i.e. the ideals $J(a, b, c) = \{e : e_x \geq a \text{ or } e_y \geq b \text{ or } e_z \geq c\}$ with $(a, b, c) \in (\{0\} \cup \mathbb{N}_{\geq 1})^3$, 0 meaning the variable is absent. An exhaustive enumeration of all $6^3 = 216$ such ideals inside the box $[0, 4]^3$ (script `reproduce.py`, checks 5b–5d) shows that no single $J(a, b, c)$ equals $(xyz) \cap [0, 4]^3$ and no intersection $J_1 \cap J_2$ of two of them does, whereas $J(1, 0, 0) \cap J(0, 1, 0) \cap J(0, 0, 1) = (xyz)$. Since every global decomposition restricts to the box, the minimal number of components is exactly 3. (Conceptually: a principal squarefree monomial (m) has the irredundant irreducible decomposition $(m) = \bigcap_{x_i | m} (x_i)$ with $|\text{supp}(m)|$ components; here $|\text{supp}(xyz)| = 3$.) $\square$

Theorem 7. $\beta_0((xyz)) = 1$.

Proof. (xyz) is generated by the single monomial xyz , and it is not the zero ideal since $xyz \in (xyz)$; the minimal monomial generating set is $\{xyz\}$ (also confirmed by brute force, `reproduce.py` check 3). $\square$

Theorem 8 (Main). $\iota((xyz)) = 3 > 1 = \beta_0((xyz))$. Consequently TLMC conjecture 00000000548 is **false**.

Proof. Immediate from Theorems 6 and 7: the conjectured inequality $\iota(I) \leq \beta_0(I)$ fails for $I = (xyz)$ in $k[x, y, z]$, i.e. at $n = 3$. $\square$

3 Remarks on the boundary

Remark 9. The counterexample lies inside the range the conjecture declares verified ($n = 3 \leq 12$), and (xyz) is itself a complete intersection (one generator), so the rider “tightness attained by complete intersections” fails as well: a complete intersection can have $\iota = 3 \neq 1 = \beta_0$.

Remark 10. For the squarefree principal monomial $(x_1 x_2 \cdots x_t)$ one has $\iota = t$ and $\beta_0 = 1$. Hence the inequality fails for every $t \geq 2$ and ι/β_0 is unbounded already on principal ideals: no repair of the form $\iota \leq \beta_0$ is possible on any class containing these examples.

Remark 11. The correct combinatorial picture is that $\iota(I)$ of a monomial ideal equals its number of maximal standard pairs, an invariant not controlled by the number of generators. Any corrected statement must bound $\iota(I)$ by a different quantity (e.g. the number of facets of an associated simplicial complex).

4 Verification artifacts

- `reproduce.py` recomputes every number from scratch by finite enumeration in the box $[0, 4]^3$: set equality, irredundancy witnesses, $\beta_0 = 1$ by brute force over generator sets, irreducibility of the three components by exhaustive search over all 252 box super-ideals of each, and minimality of the component count over all 216 irreducible-form ideals. Output: $\iota = 3$, $\beta_0 = 1$.
- `lean4/` contains an axiom-free Lean 4 formalization (toolchain `leanprover/lean4:v4.33.1`, no Mathlib). `Main.lean` proves the decomposition $(xyz) = (x) \cap (y) \cap (z)$ (pointwise, by case analysis), irredundancy via the witnesses yz, xz, xy , the irreducibility lemma (constructively, using $u + v$ instead of uv to avoid case analysis on natural numbers), $\beta_0 = 1$, and the numeric conclusion $3 > 1$. `Check.lean` audits all theorems with `#print axioms`: every one reports *does not depend on any axioms*.
- `build/main.pdf` is this document; `tectonic main.tex --outdir build` rebuilds it.